

Project ID :**25 - 26J - 320**

1. Topic (12 words max)

Digital Platform for Social Welfare Services Management in Sri Lanka

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Sri Lanka offers a wide range of welfare programs such as Samurdhi, disability allowances, elderly pensions, and nutritional aid. However, the current welfare system suffers from serious inefficiencies, such as manual application handling, fragmented data, lack of real-time tracking, and delays in approval processes. These challenges result in ineligible beneficiaries receiving aid while genuinely deserving citizens often face barriers to access.

In an era where e-governance and digital transformation have shown success globally especially in countries like India and Bangladesh, Sri Lanka still lacks a unified, citizen-centric digital solution to manage welfare services. Moreover, the absence of real-time data collection and intelligent decision support tools hinders effective policy formation and resource allocation.

This research aims to address these systemic gaps through a centralized, AI-powered digital platform designed for citizens, Divisional Secretariat officers, and policymakers. The platform will integrate smart eligibility prediction, AI-based priority handling, IoT-based multilingual chatbot assistance to enhance transparency, efficiency, and accessibility in the welfare ecosystem.

The proposed system not only automates and streamlines welfare distribution but also utilizes predictive analytics and machine learning to support proactive, data-driven decision-making ensuring aid reaches those who need it most.

* Ahmed, M. U. (2019) - South Asian Journal of Policy and Governance.

* United Nations (2022) - E-Government Survey

* ICTA Sri Lanka (2023) - Digital Government Strategy

* Bhatnagar, S. (2004) - E-Government Book (SAGE)

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a web-based and mobile accessible digital platform incorporating modern AI technologies to transform social welfare delivery. The system is composed of three modules: Citizen Portal, Divisional Secretariat Officer Dashboard, and Policy Admin Panel.

For **citizens**, the platform provides a smart eligibility checker using AI, a chatbot for multilingual support, real-time application tracking, document uploads system, and appointment scheduling features.

For **government officers**, it offers a centralized dashboard to process and review applications, validate documents, and reduce delays in manual verifications.

For **policymakers**, the solution provides AI-powered dashboards with real-time visualizations of welfare usage, fraud detection, priority risk maps, and automated insights for better decision-making.

Key AI components include:

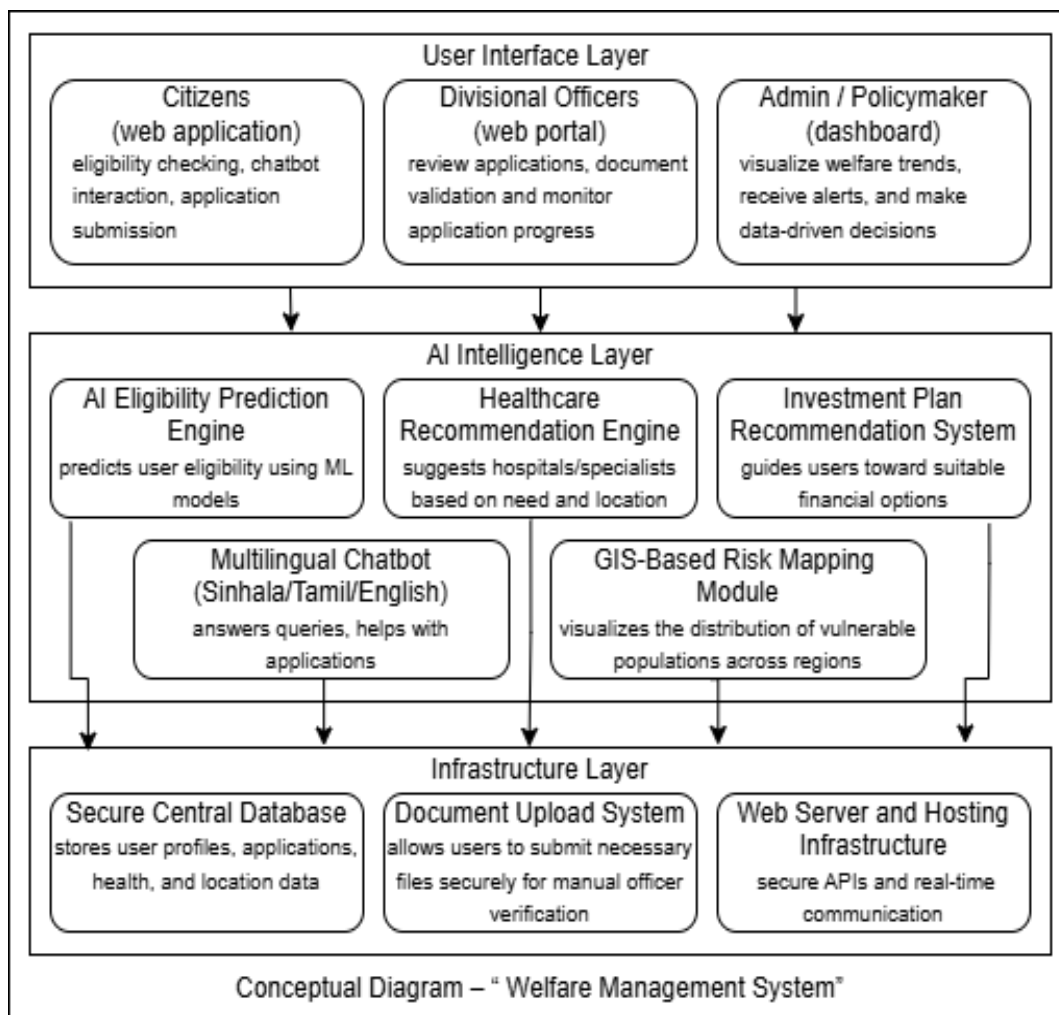
Eligibility Prediction Engine and Healthcare Recommendation System

Multilingual Chatbot

Investment Plan Recommendation Module

AI + GIS Mapping System

This integrated ecosystem increases service efficiency, ensures better inclusivity, and supports predictive governance.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project leverages specialized expertise across multiple AI domains to digitally transform Sri Lanka's social welfare services. Each system component is designed to solve a specific societal challenge through domain-aligned techniques, requiring targeted datasets and specialized knowledge.

- **Eligibility Prediction Engine & Healthcare Recommendation System**

This dual-module applies machine learning and health informatics to improve citizen access to services. The prediction engine uses classification models (e.g., decision trees, logistic regression) trained on socio-economic and demographic data to evaluate program eligibility. The healthcare recommendation tool suggests hospitals and specialists based on user health profiles, leveraging ranking algorithms and geographic filters. Both systems require labeled datasets including income levels, age, household size, health status, and medical facility data.

- **Multilingual Conversational Assistance System**

This Natural Language Processing (NLP)-based chatbot supports Sinhala, Tamil, and English to assist users with application tracking, documentation, and eligibility queries. It utilizes dialogue management, intent recognition, and named entity extraction, developed using frameworks like Rasa and spaCy. It requires multilingual corpora, annotated intent datasets, and domain-specific FAQs in all supported languages.

- **Investment Plan Recommendation Module**

This module recommends suitable financial support and micro-investment options to low-income users using clustering and collaborative filtering techniques. It aligns with financial AI applications and requires anonymized user income data, expense patterns, and government/NGO financial aid schemes.

- **AI-GIS Welfare Insights Platform**

This system uses geospatial AI and predictive analytics to identify welfare-demand hotspots (e.g., areas with high concentrations of chronic patients, pregnant mothers). It employs spatial data modeling and heatmap visualizations to support policy-level decision-making. Required data includes district-level health indicators, welfare distribution data, and population demographics.

All data will be handled ethically and in compliance with national data protection laws.

8.Objectives and Novelty

Main Objective To design and develop an AI-integrated digital welfare platform to improve efficiency, transparency, and accessibility in Sri Lanka's social protection programs.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Manawadu G.M. S. T IT22926494	Design AI eligibility prediction model and healthcare recommendation system	Data collection, model training, hospital/specialist recommendation engine development	AI-powered welfare and healthcare assistance tailored to citizen profiles
Shamika A. G IT22585080	Develop multilingual chatbot	Design and integrate chatbot in Sinhala, Tamil, and English	First-ever tri-lingual chatbot for Sri Lankan welfare services
Dissanayake D.M.V.S IT22366818	Build AI-based financial guidance module	Build system to recommend suitable investment or savings plans based on user profiles	Smart financial empowerment system for low-income welfare recipients
Wickramasinghe R P Y N IT22214584	Develop AI + GIS mapping for vulnerable groups	Integrate health and location data to map chronic patients, pregnant women, and disabled populations	Predictive geo-mapping for targeted resource allocation and policy insights

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Manawadu G.M. S. T IT22926494	Specializing in Machine Learning and Health Informatics , Shani is responsible for two critical AI components. First, she develops the Eligibility Prediction Engine , which uses classification models to assess whether a citizen qualifies for specific welfare programs based on demographic and socio-economic data. Second, she builds the AI-based Healthcare Recommendation System , which analyzes user health needs (e.g., chronic illness, pregnancy) and suggests the most suitable hospitals and specialist doctors in Sri Lanka. These tasks involve working with healthcare datasets, ML algorithms, and ranking systems. All aligned with her specialization in AI for social good and health systems.
Shamika A. G IT22585080	With a specialization in Natural Language Processing (NLP) and Conversational AI , this member is developing the Multilingual Chatbot . The chatbot will support Sinhala, Tamil, and English, helping users get real-time assistance about application status, documentation, and eligibility. The implementation includes dialogue management, intent recognition, and multi-language support using NLP tools such as Rasa, spaCy, or pre-trained language models. This role directly matches the members' expertise in building inclusive, user-friendly conversational systems.
Dissanayake D.M.V.S IT22366818	This member focuses on AI in Financial Technologies . They are responsible for creating the AI-based Investment Plan Recommendation Module , which helps users especially low-income citizens identify suitable and safe micro investment or financial aid schemes based on their financial background and risk profile. Using clustering techniques and recommender system logic, the component aligns with their specialization in applying AI for financial decision support.
Wickramasinghe R P Y N IT22214584	With expertise in Geospatial AI and Predictive Analytics , this member leads the AI + GIS Mapping System to identify regions in Sri Lanka with high concentrations of disabled individuals, pregnant women, or chronic patients (e.g., kidney disease). This uses spatial data analysis and heatmap visualization to support data-driven policy decisions and resource allocation. The work reflects a strong alignment with geo-AI specialization and social policy optimization.

8. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Senior Lecturer	Anjalie	Gamage	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

b) Does the proposed topic exhibit novelty?

c) Do you believe they have the capability to successfully execute the proposed project?

d) Do the proposed sub-objectives reflect the students' areas of specialization?

e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.